

Macroeconomics – Try-Out Exam Answers

AY 2025/2026 | Macroeconomics (MBA Bridging Program)

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PART I – Case: Impact of Changes in Electricity Prices

Question 1.1 – Initial Long-Run Equilibrium (Time 0) (0.75 p)

In **long-run equilibrium**, actual prices equal expected prices and output equals the natural rate (Y^*). All three curves intersect at a single point (P_0, Y^*).

- **LRAS** – vertical at Y^* (supply determined by real factors only)
- **SRAS₀** – upward-sloping (sticky wages in the short run)
- **AD₀** – downward-sloping (wealth + interest-rate effects)

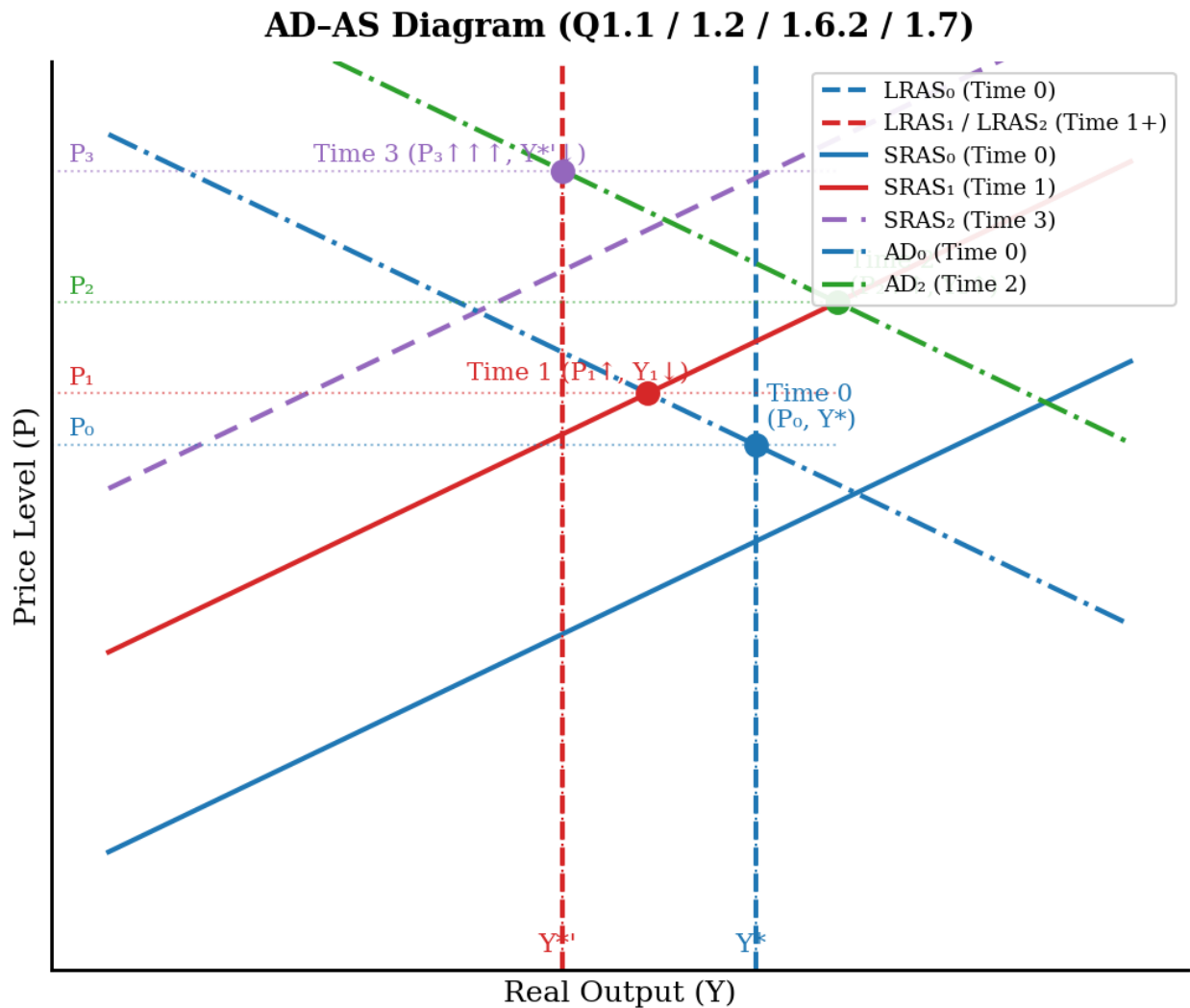


Figure 1: AD-AS diagram showing all four equilibrium positions (Time 0–3)

Time 0 is the intersection of all three curves at (P_0, Y^*).

Question 1.2 – Short-Run Impact of the Breakdown (Time 1) (1.5 p)

A power-station breakdown is a **permanent adverse supply shock**: electricity is a crucial production input across all industries.

Mechanism:

1. Higher electricity prices → **production costs rise** for firms economy-wide.
2. Firms respond by **reducing output** and **raising prices**.
3. The **SRAS curve shifts LEFT/UPWARD** ($SRAS_0 \rightarrow SRAS_1$) — firms supply less at every price level.
4. Because the breakdown is *permanent*, **productive capacity is permanently reduced** → **LRAS also shifts left** ($Y^* \rightarrow Y^*$).
5. AD_0 is unchanged initially (no demand-side shock yet).

Short-run result (Time 1):

Variable	Direction
Output (Y)	Falls ($Y_1 < Y^*$)
Price level (P)	Rises ($P_1 > P_0$)
Unemployment	Rises

This combination of falling output and rising prices is called **stagflation**.

(See diagram above — Time 1 point is the intersection of AD_0 and $SRAS_1$.)

Question 1.4 – Short-Run Impact on the Phillips Curve (Time 0 → Time 1) (1.5 p)

Key points on the diagram:

- $LRPC_0$ – vertical at the initial natural unemployment rate (u^*)
- $LRPC_1$ – shifts **RIGHT** to u^* because the permanent breakdown raises the natural unemployment rate (lower productive capacity → fewer jobs in the long run)
- $SRPC_0$ – initial short-run Phillips curve (Time 0 at u^* , π_0)
- $SRPC_1$ – shifts **upward** due to the supply shock: firms face higher costs, raise prices → higher inflation at every level of unemployment
- **Time 1 (SR)** – the economy moves up and to the right along $SRPC_1$: **both inflation and unemployment increase** simultaneously (stagflation in Phillips curve terms)

The supply shock is **unfavorable**: policymakers cannot reduce both inflation and unemployment at the same time. Accepting higher inflation does NOT reduce unemployment here (unlike a standard demand expansion).

Question 1.5 – Labour Market with Binding Minimum Wage (Time 0 → Time 1) (2 p)

Initial situation (Time 0):

- The minimum wage $W_{\min}$ is **above** the market-clearing wage → labour supply exceeds labour demand at $W_{\min}$ → **structural unemployment** already exists (gap between L_S and L_0).

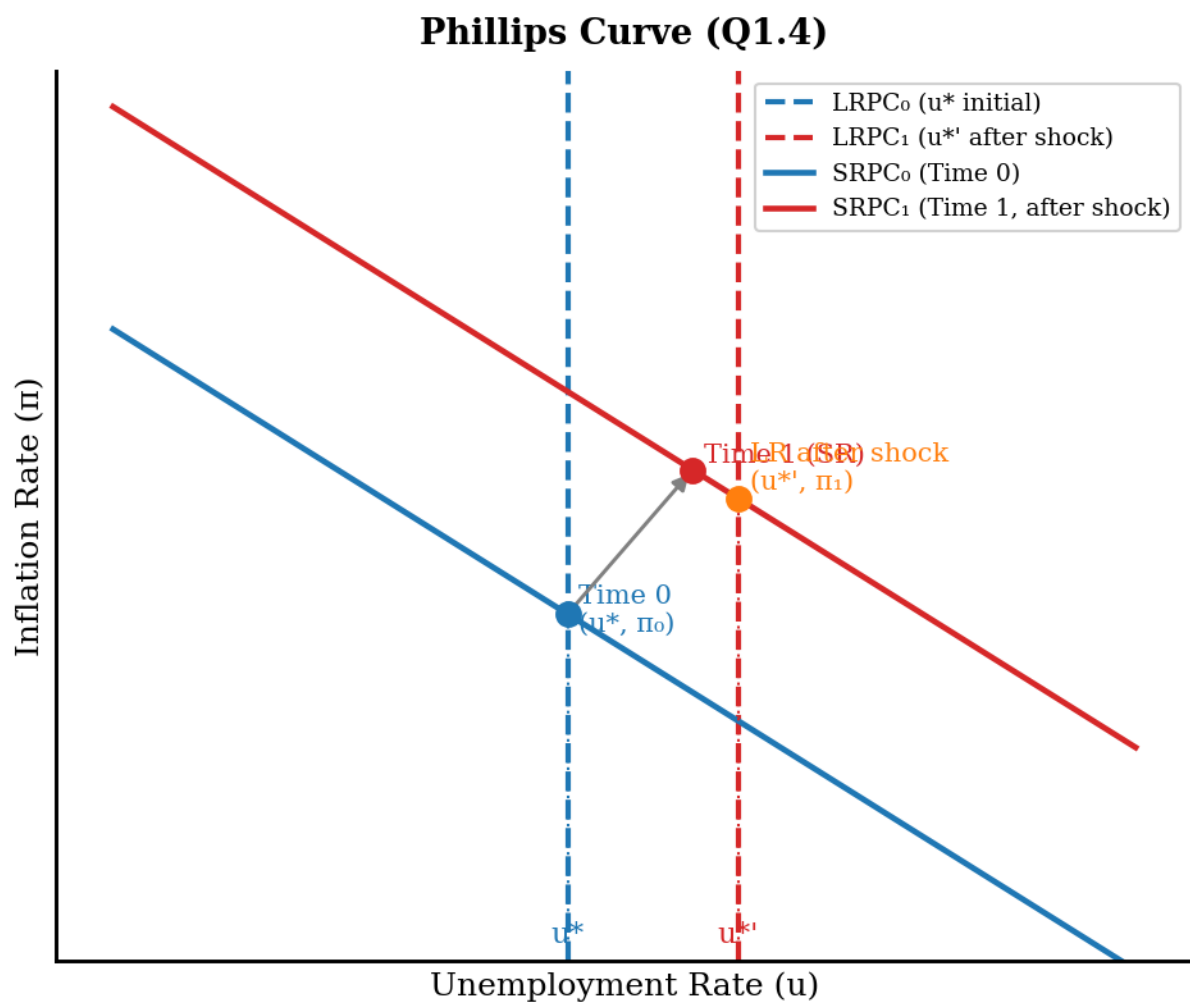


Figure 2: Phillips Curve diagram showing supply shock shift

Labour Market with Binding Minimum Wage (Q1.5)

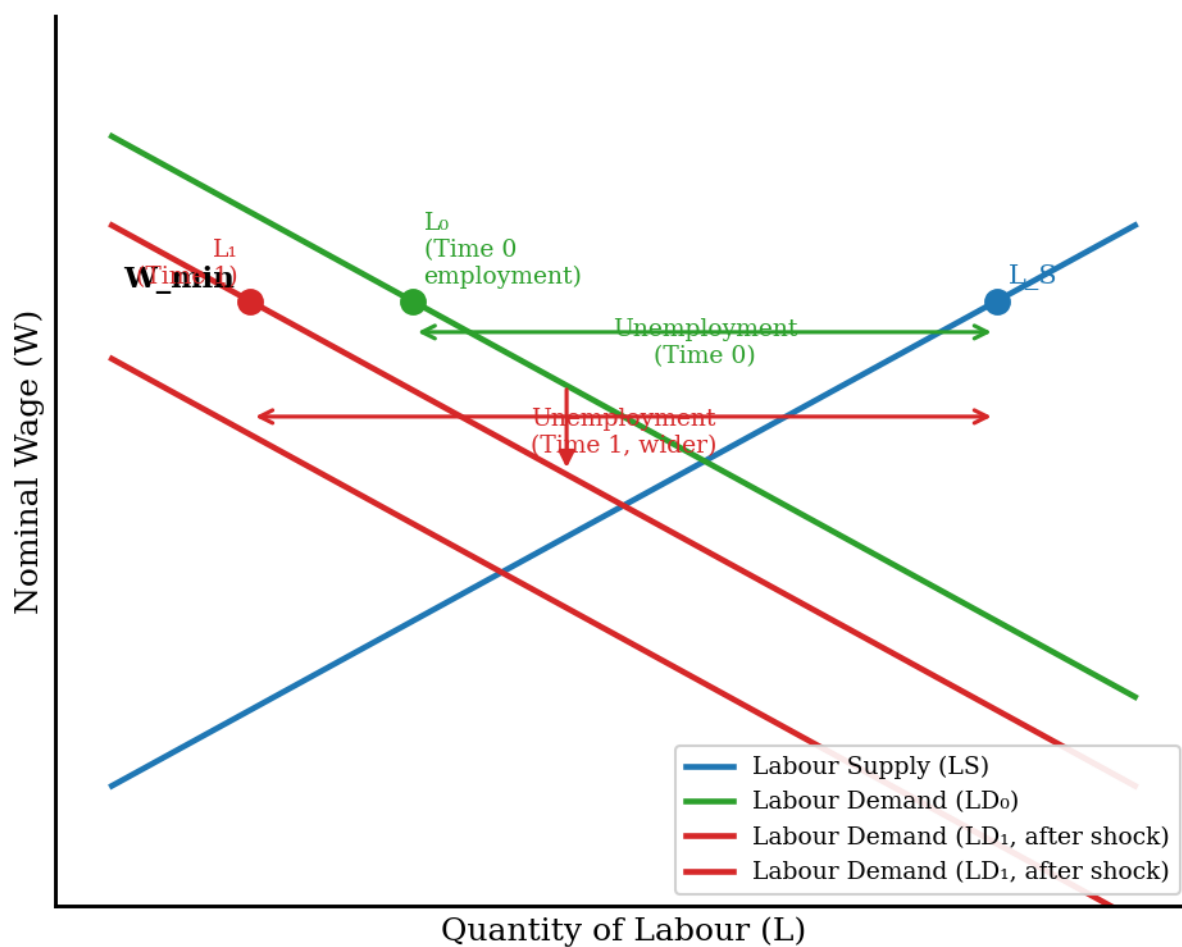


Figure 3: Labour market with binding minimum wage

After the breakdown (Time 1):

- Higher electricity costs reduce firm profitability → **firms reduce labour demand** → LD_0 shifts LEFT to LD_1 .
- Wages **cannot fall below $W_{\min}$** (legally binding floor) — the normal market-adjustment mechanism is blocked.
- At $W_{\min}$, firms now demand even fewer workers: $L_1 < L_0$.
- Labour supply (L_S) at $W_{\min}$ is unchanged.
- **The unemployment gap widens:** $(L_S - L_1) > (L_S - L_0)$.

Conclusion: The binding minimum wage prevents wages from falling to absorb the shock, so the entire adjustment falls on **employment** — unemployment rises more than it would in a flexible wage market.

Question 1.6 – Government Policy Response

1.6.1 – Type of Policy (0.75 p)

To keep economic activity high before elections, the government will implement:

Expansionary Fiscal Policy — specifically an **increase in government spending ($\uparrow G$)** and/or a **cut in taxes ($\downarrow T$)**.

This directly or indirectly increases aggregate demand, partially offsetting the recession caused by the supply shock.

1.6.2 – Impact on the AD–AS Diagram (Time 2) (0.75 p)

(See the AD–AS diagram above — the green AD_2 curve and the green “Time 2” point.)

- Expansionary fiscal policy shifts **AD to the right** ($AD_0 \rightarrow AD_2$).
- Moving along $SRAS_1$, the new short-run equilibrium (Time 2) shows:
 - **Output rises** ($Y_2 > Y_1$, recovering toward Y^*)
 - **Price level rises further** ($P_2 > P_1$) — the stimulus adds additional inflationary pressure on top of the supply shock.

1.6.3 – Effectiveness Depending on Interest Sensitivity of Investment (IS-LM) (2 p)

Fiscal policy ($\uparrow G$) shifts the **IS curve to the right**. Higher income raises money demand → interest rate (i) rises → **private investment is crowded out**. The degree of crowding-out depends on how sensitive investment is to the interest rate.

Sensitivity	IS slope	IS shift result	Crowding-out	Fiscal effectiveness
High	Flat	Small ΔY , large Δi	Large	Less effective
Low	Steep	Large ΔY , small Δi	Small	More effective

Conclusion: Fiscal policy is **most effective when the interest sensitivity of investment is LOW** (steep IS curve). In this case, even though the interest rate rises, investment barely responds, so very little of the fiscal stimulus is crowded out.

IS-LM: Effectiveness of Fiscal Policy by Interest Sensitivity of Investment (Q1.6.3)

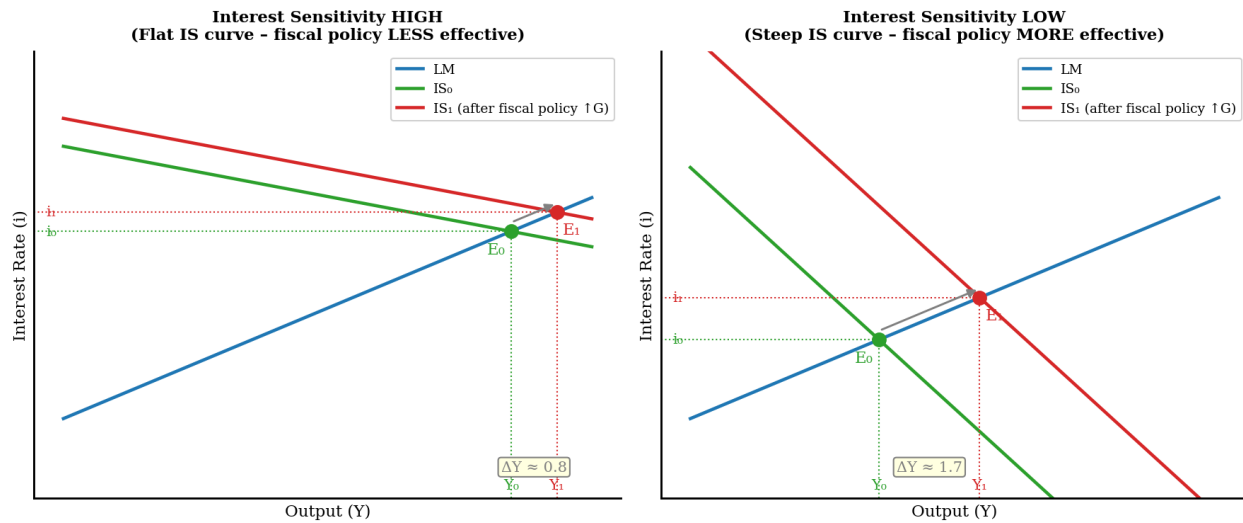


Figure 4: IS-LM diagrams comparing high vs low interest sensitivity

Question 1.7 – Long-Run Outcome (Time 3) (0.75 p)

(See AD-AS diagram above — the purple “Time 3” point.)

Since the breakdown is **permanent**, the long-run adjustment unfolds as follows:

1. The economy is above the new natural rate at Time 2 (output $Y_2 > Y^{*}$ and prices are rising).
2. Over time, workers and unions **update their inflation expectations upward** → they bargain for higher wages → firms face even higher costs → **SRAS continues to shift left** ($SRAS_1 \rightarrow SRAS_2$) until it passes through the new LRAS (at Y^{*}).
3. Long-run equilibrium (Time 3) is where **AD₂ intersects the new LRAS** (at Y^{*}):
 - **Output = Y^{*}** (permanently lower than original Y^{*})
 - **Price level = P_3** (significantly higher than P_0 ; government stimulus made it even higher)

The government's expansionary policy **only raises prices in the long run** — it cannot permanently move output above the new natural rate Y^{*} . The long-run sacrifice is both lower output (structural) and higher inflation (policy-induced).

PART II – Multiple Choice

Answer Sheet

Question	A	B	C	D
2.1	✓			
2.2				✓
2.3			✓	
2.4			✓	
2.5			✓	

Question 2.1 – Money Market → Answer: A (Both statements are correct)

The diagram has **value of money (1/P)** on the vertical axis and **nominal quantity of money (M)** on the horizontal axis. The money supply (MS) is a vertical line controlled by the central bank; money demand (MD) is downward-sloping.

Statement 1 – A → D via Quantitative Easing: ✓ Correct

Quantitative easing (QE) = the central bank purchases long-term assets from banks, **increasing the money supply**. The MS curve shifts **rightward**. The new equilibrium has *more money* and a *lower value of money* (higher price level). If D lies to the right and below A on the MD curve, this movement is consistent with QE.

Statement 2 – D → C via increased inclination to hold cash over demand deposits: ✓ Correct

When households shift holdings from demand deposits to cash, the **currency-deposit ratio rises** → the **money multiplier falls** → the effective money supply M1 (currency + demand deposits) **contracts** → MS shifts **leftward**. The equilibrium moves to a point with *less money* and a *higher value of money*. The movement D → C (upper-left direction) is consistent with this effect.

Answer: A

Question 2.2 – Labour Market Statistics → Answer: D

Calculations:

	2014	2015
Working-age population (18–65)	10,000,000	11,000,000 (+1M)
Employed	5,000,000	5,000,000 (unchanged)

	2014	2015
Unemployed	1,000,000	2,000,000 (doubled)
Labour force	6,000,000	7,000,000
Not in labour force	4,000,000	4,000,000
Unemployment rate	$1/6 = 16.67\%$	$2/7 = 28.57\%$
Employment rate	$5/10 = 50.00\%$	$5/11 = 45.45\%$
Labour Force Participation Rate	$6/10 = 60.00\%$	$7/11 = 63.64\%$

Checking each statement:

- **A)** Unemployment rate doubled? → doubled would be 33.33%; actual = 28.57% → **FALSE**
- **B)** Employment rate unchanged? → $50\% \neq 45.45\%$ → **FALSE**
- **C)** People not in labour force decreased? → $4M = 4M$ (unchanged) → **FALSE**
- **D)** LFPR increased? → $60.00\% \rightarrow 63.64\%$ → **TRUE ✓**

Answer: D

Question 2.3 – Before-Tax Nominal Interest Rate → Answer: C ($15\% < x \leq 20\%$)

Given:

Variable	Value
After-tax real interest rate	12%
Inflation rate	2%
Tax rate	12.5%

Step 1 – Find after-tax nominal interest rate using the Fisher equation:

$$\text{After-tax nominal rate} = \text{After-tax real rate} + \text{Inflation} = 12\% + 2\% = 14\%$$

Step 2 – Relate after-tax nominal rate to before-tax nominal rate:

$$\text{After-tax nominal rate} = x \times (1 - \text{tax rate})$$

$$14\% = x \times (1 - 0.125) = 0.875x$$

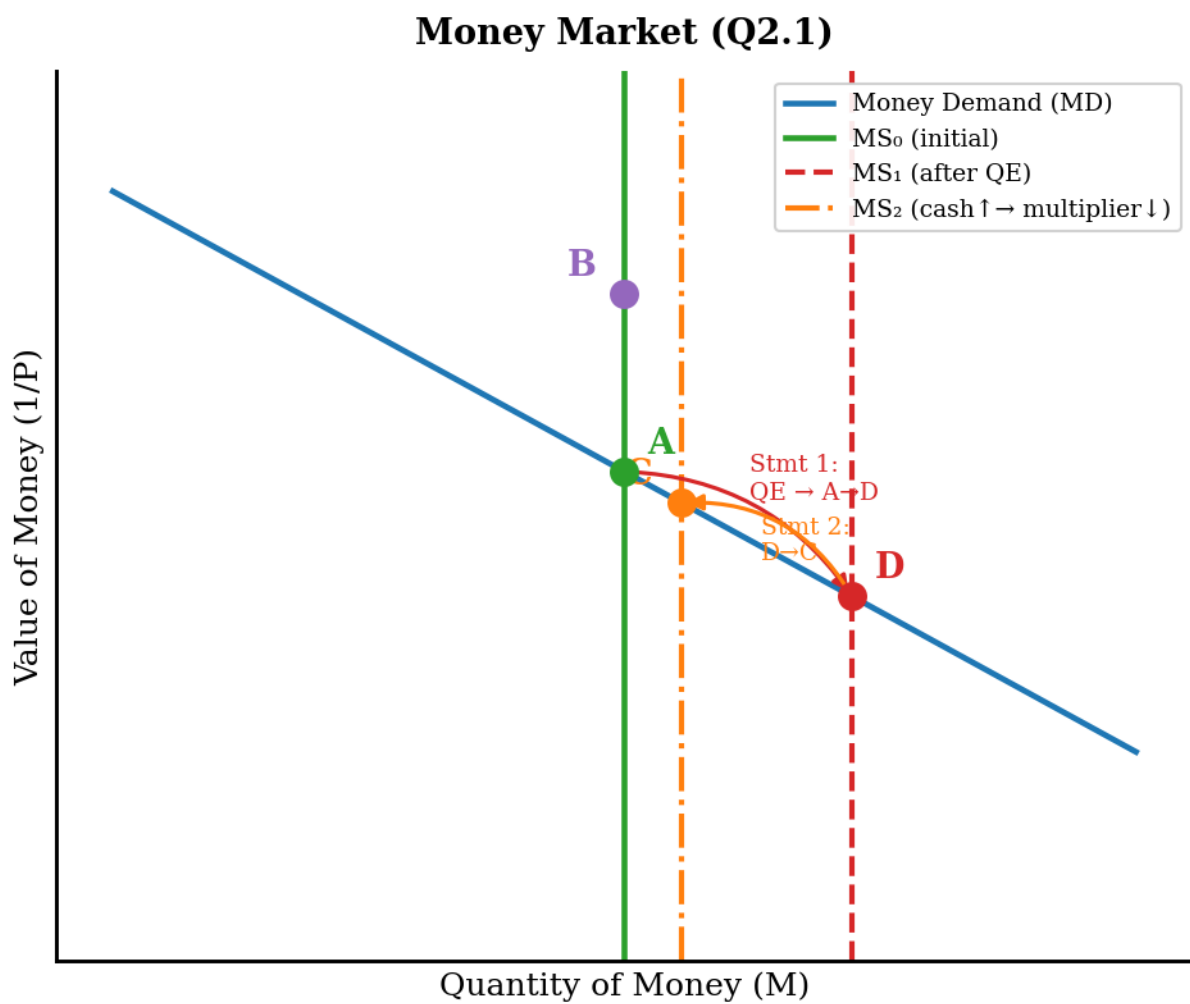


Figure 5: Money Market diagram

Step 3 – Solve for x:

$$x = \frac{14\%}{0.875} = \mathbf{16\%}$$

Since $15\% < 16\% \leq 20\%$:

Answer: C

Question 2.4 – Trading Partner Growth → Answer: C (Only Statement 2 correct)

Context: Country X's main trading partner experiences strong growth → that country's income rises → it imports more from Country X → **Country X's net exports (NX) increase** → IS curve shifts right → AD increases.

Statement 1: "Assume $I = f(i, Y)$. The short-run effect of this event is theoretically ambiguous." → **INCORRECT**

With $I = f(i, Y)$ (investment depends on both the interest rate and income): - $NX \uparrow$ → IS shifts right → in IS-LM equilibrium: both **Y rises and i rises** - The effect on investment *alone* is ambiguous ($Y \uparrow$ raises I, but $i \uparrow$ lowers I) - However, the effect on **total output (Y)** is **unambiguously positive** — the IS shift rightward with an upward-sloping LM always yields a higher Y - The Keynesian stability condition ($\Delta Z / \Delta Y < 1$) ensures a well-defined, higher equilibrium

Statement 1 is **wrong**: the short-run effect on aggregate output is NOT ambiguous.

Statement 2: "In the long run, the price level will increase more without automatic stabilizers." → **CORRECT**

Automatic stabilizers (progressive taxation, unemployment benefits) **dampen** the AD increase: - As income rises → taxes automatically increase, transfers decrease → AD shift is **smaller** - Without stabilizers → the full multiplier plays out → **larger AD shift** → larger long-run price level increase

Since LRAS is vertical, a larger AD shift produces a larger price level increase in the long run. ✓

Answer: C

Question 2.5 – GDP Investment Component → Answer: C

The question asks which transaction affects the **Investment (I)** component of Eurozone GDP in year t.

Recall: **I = Fixed investment + Change in inventories**

Option	Analysis	Affects I?
A British firm builds factory in London	Investment, but UK is outside the Eurozone — does not affect Eurozone GDP	X
B Belgian buys German car (produced year t)	Both countries in Eurozone; car bought by consumer = Consumption (C) , not I	X

Option	Analysis	Affects I?
C Dutch company sells carpet produced in t-1	Carpet was produced in t-1 → counted as inventory investment (+I) in t-1 . Selling from inventory in year t → inventory falls → negative inventory investment (-I) in year t . This directly affects the I component.	✓
D Greek resident buys old house	Old house = existing asset, not new production; sale of existing assets does not enter current GDP	✗

Option C is the only transaction that affects the Investment (I) component of Eurozone GDP in year t — through a reduction in inventories (negative inventory investment).

Answer: C

End of Answers Document